SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1105

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models**.** *(BL4)*

Assessment Tool Weightage: 40%

QUESTIONS:4 **Total Marks:40**

1. Develop a system using UML for Passport Management System, It should verify the passport details of the applicant by the central computer. The details regarding the passport will be provided to the central computer and the computer will verify the details of the applicant and provide approval to the office. Then the passport will issue from the office to the applicant.

2. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

3. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

4. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

5. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

6. Develop a system using UML for implementing E-Ticketing System. This is to be used by the passenger for reserving the tickets for their travel. This E-ticketing is organized by the central system. The information is provided by the railway reservation system.

7. Develop a system using UML for implementing Credit Card Processing System. In this system, the
   cardholder can purchase items and pays the bill with the aid of a credit card. The cashier should accept the card and proceed with the transaction using the central system. The bill is to be verified and the items are delivered to the cardholder based on it.

8. Develop a system using UML for Software Personnel Management System. It should be used to process the details of a person who works in a software company. The details are to be stored in the central management system for cross-checking the person's details.

9. Develop a system using UML for E-Book Management System. It should be designed to manage the books that were read through the internet. That should consist of the details of the e-book that were read by the user online. It will be controlled by the central system. This system should act as a backup of all details together.

10. Develop a system using UML for Recruitment System, to recruit the applicant for a particular job in the company. It was controlled by the central management system to manage the details of the particular applicant who has to be recruited by a company.

Code of Conduct:
I <u>DARSHINIE KARANI.V.M (192520033)</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

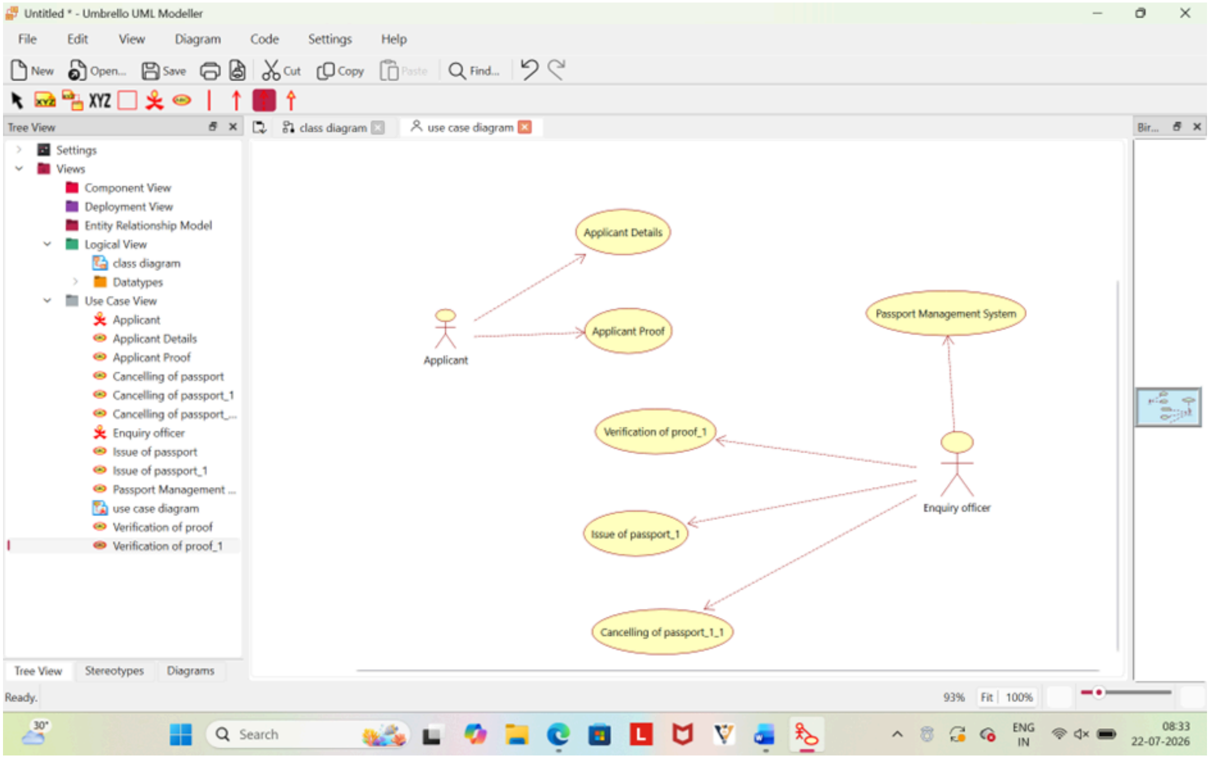
RUBRICS FOR CALCULATION:

Experiments

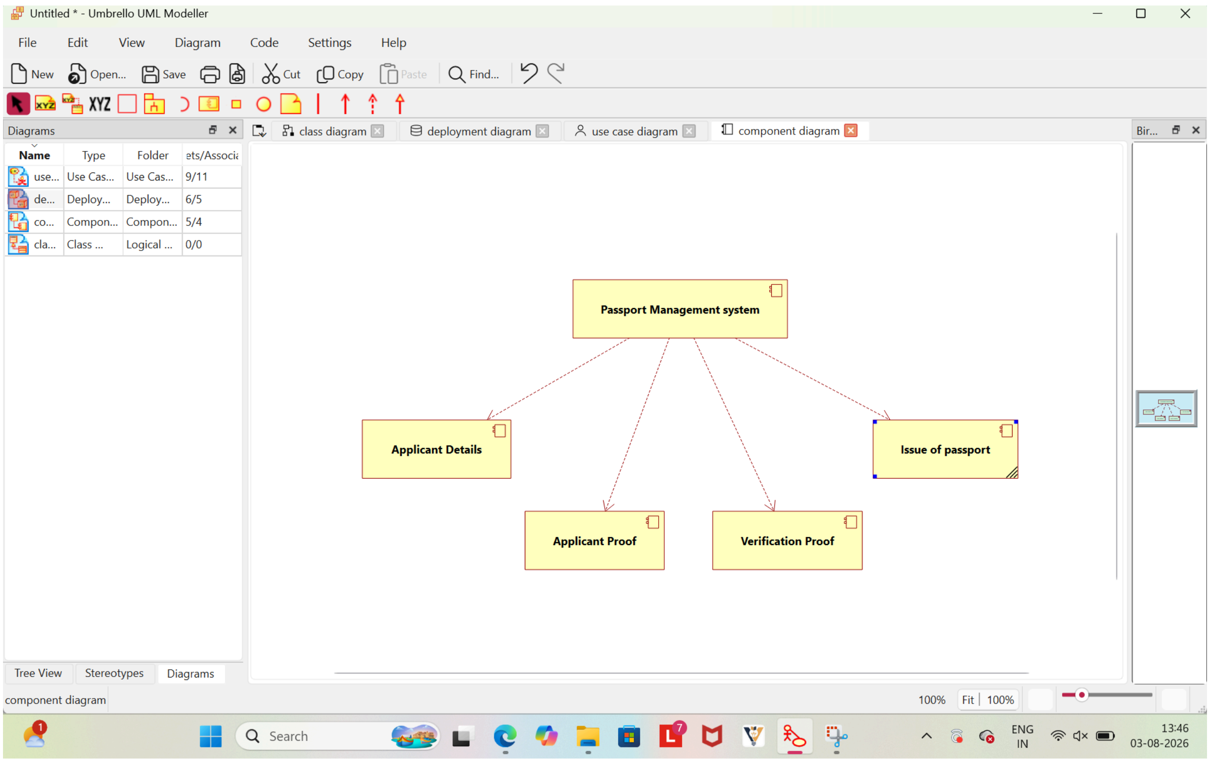
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

LAB EXPERIMENT ANSWERS

1.

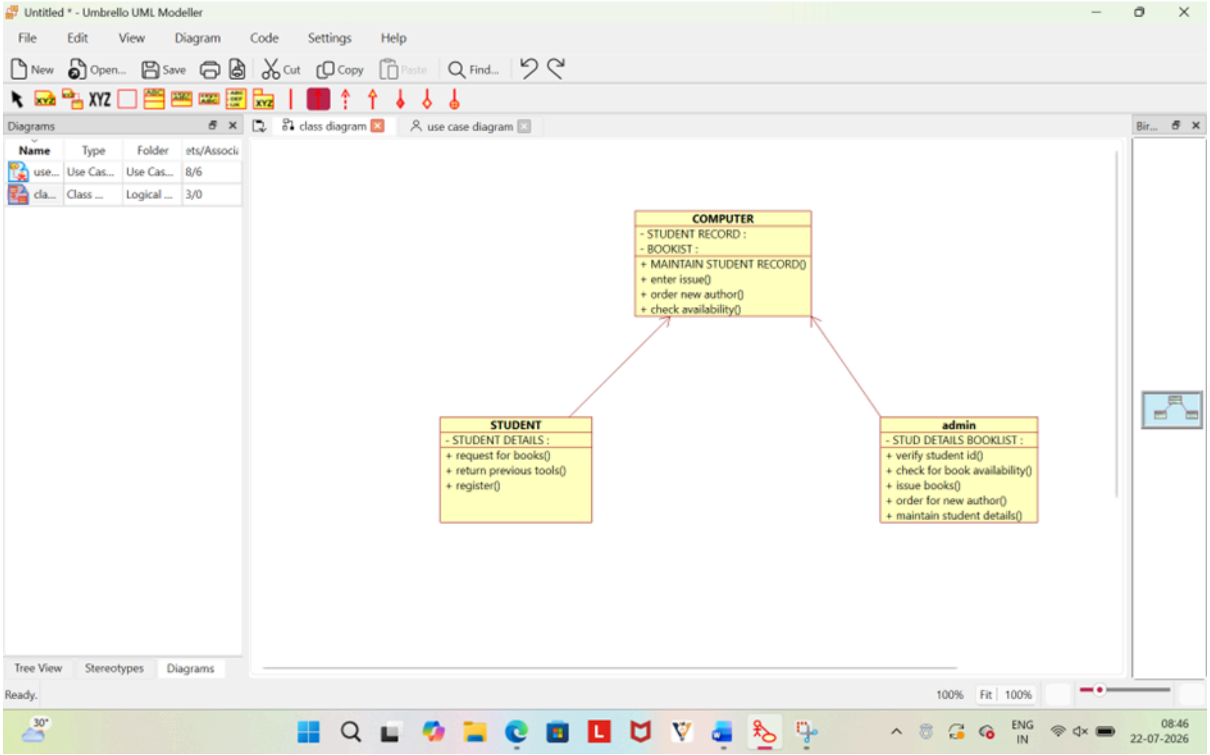


Use case diagram

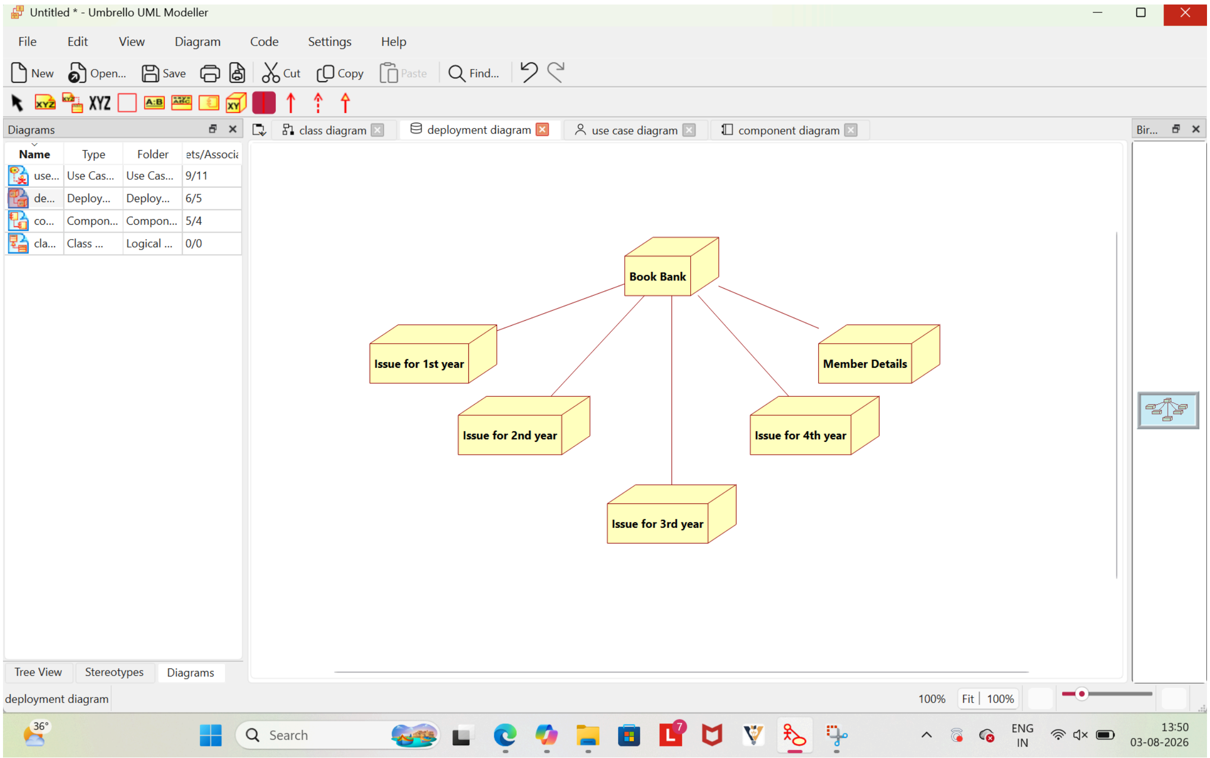


Component Diagram

2.

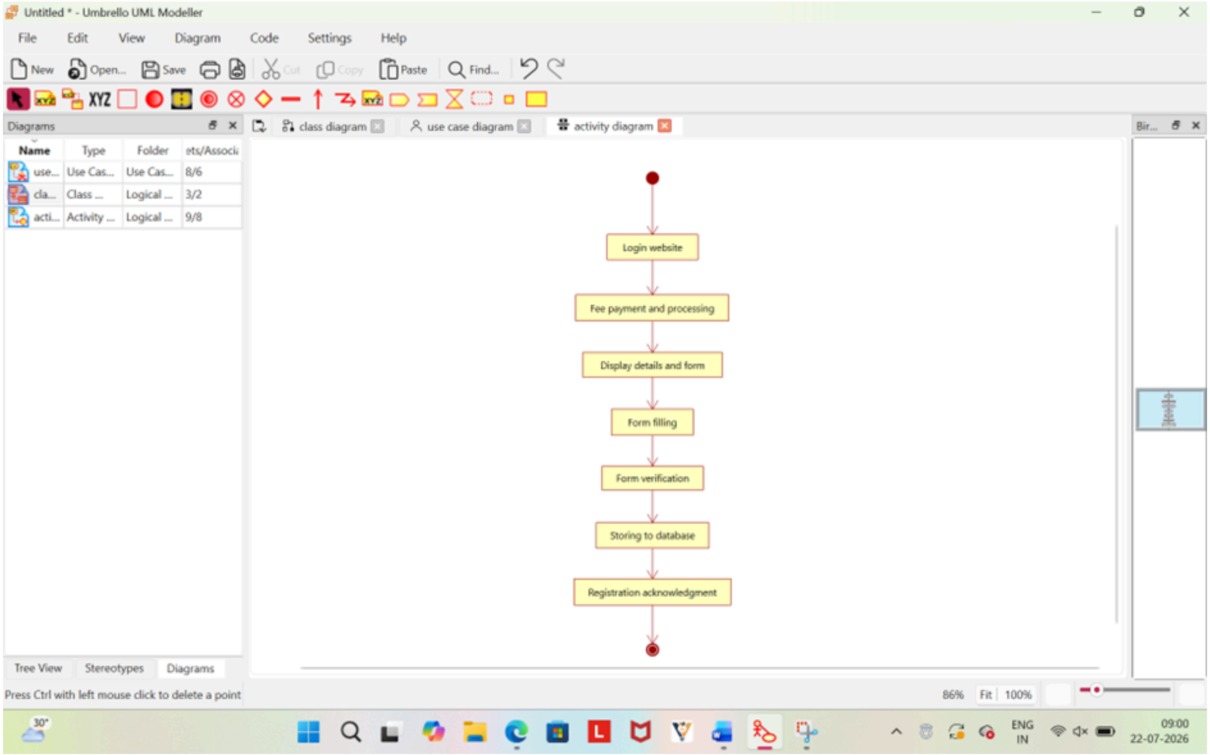


Class Diagram

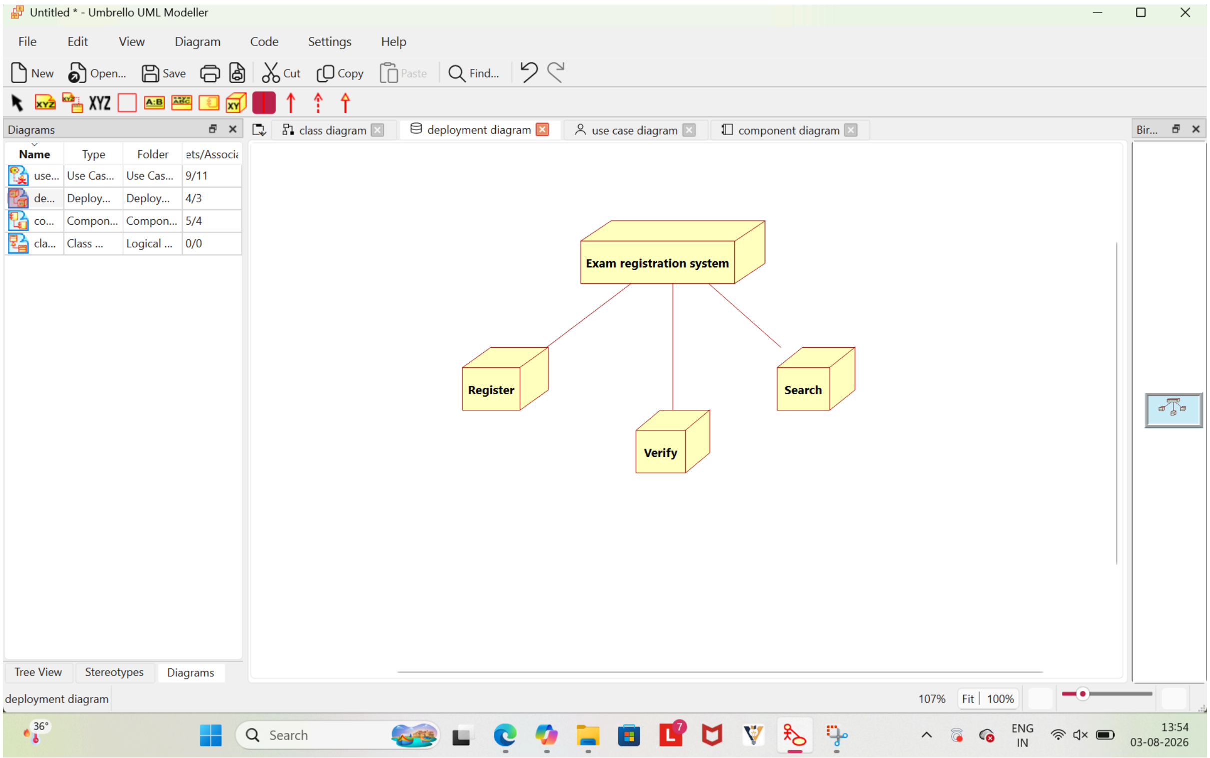


Deployment Diagram

3.



Activity Diagram



Deployment Diagram

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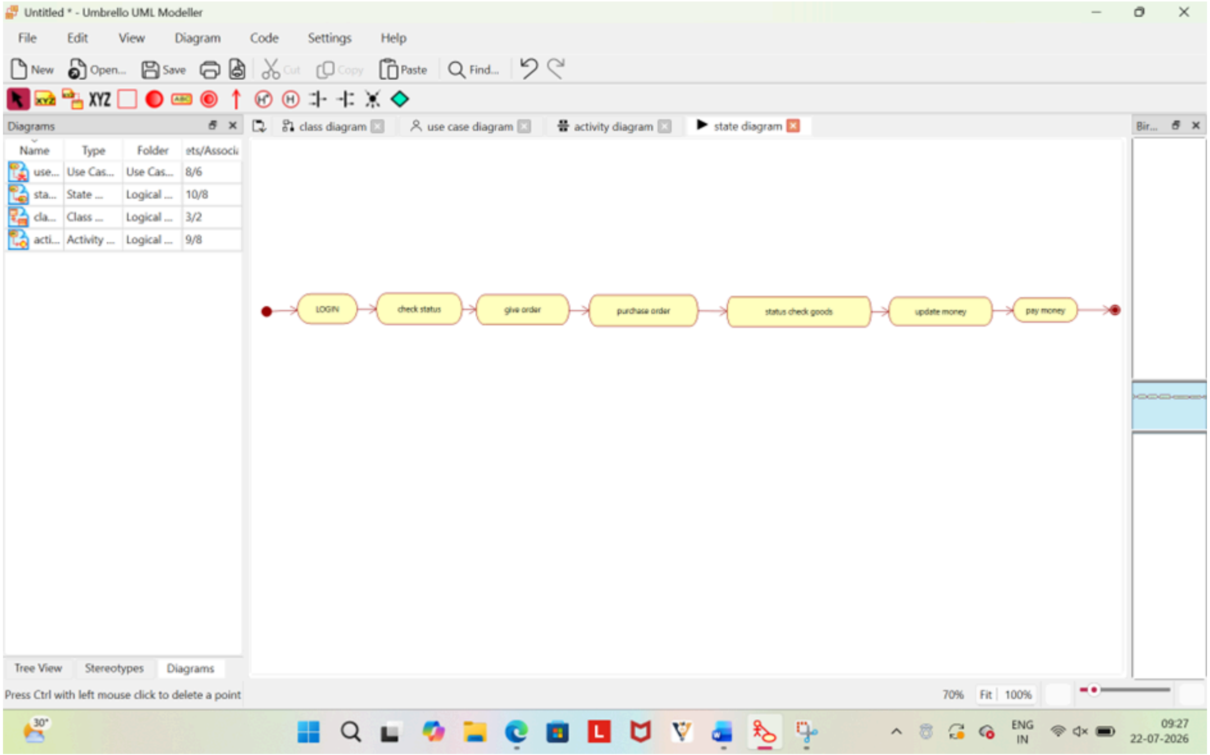
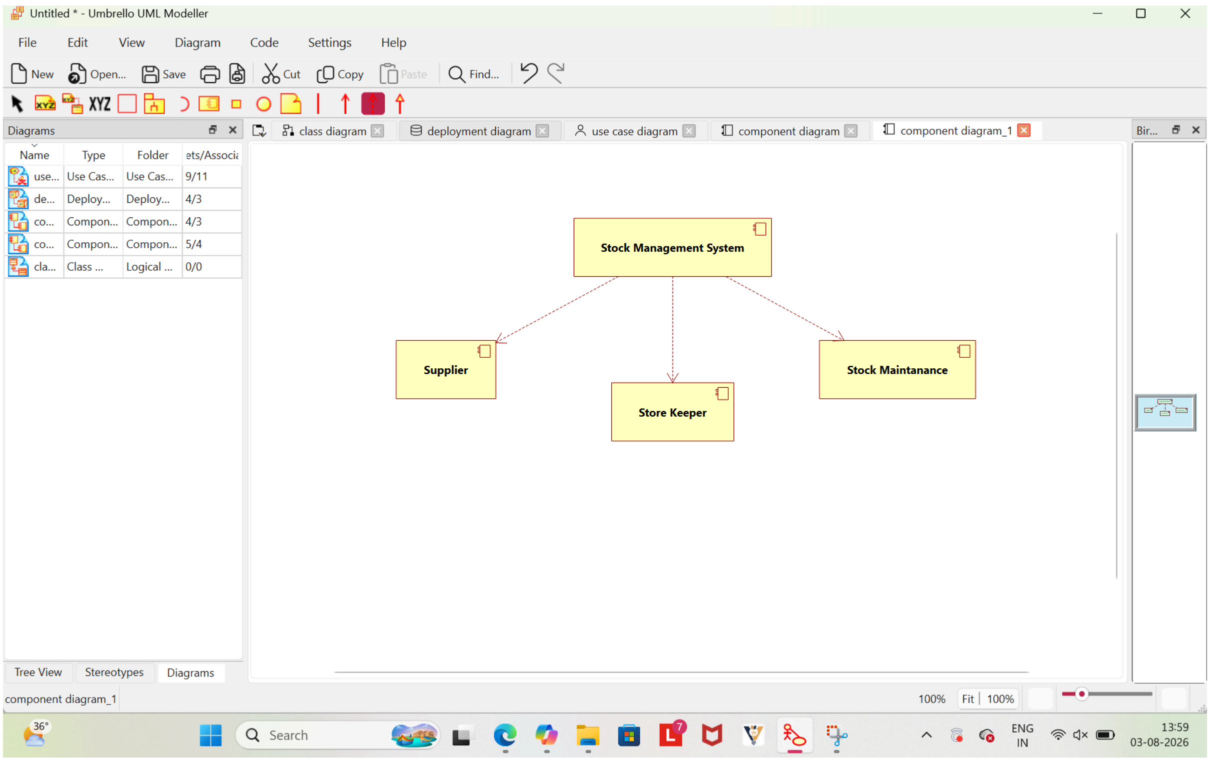
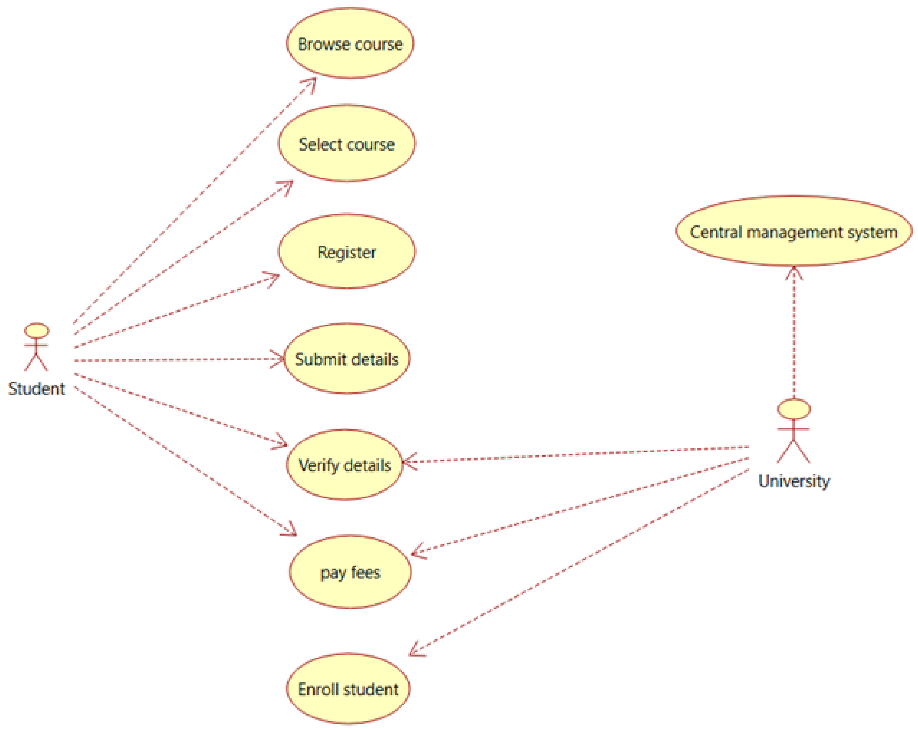


State Diagram

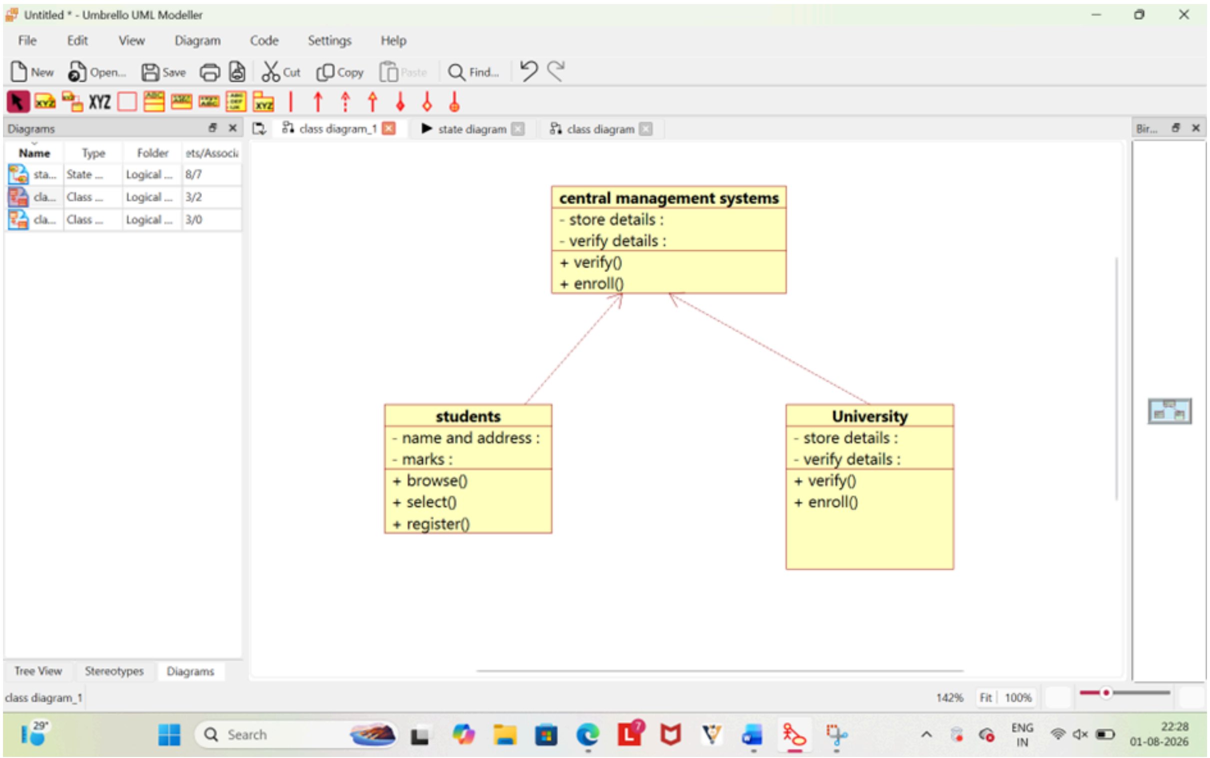


Component Diagram

5.

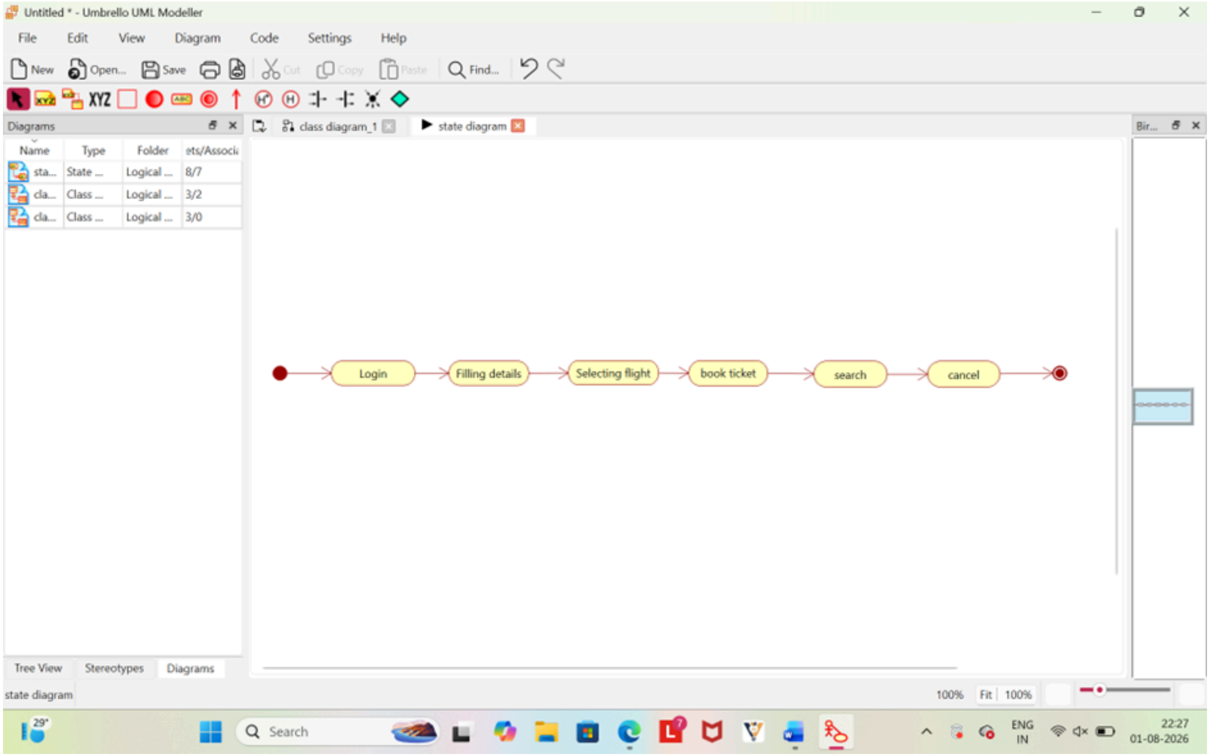


Use case Diagram

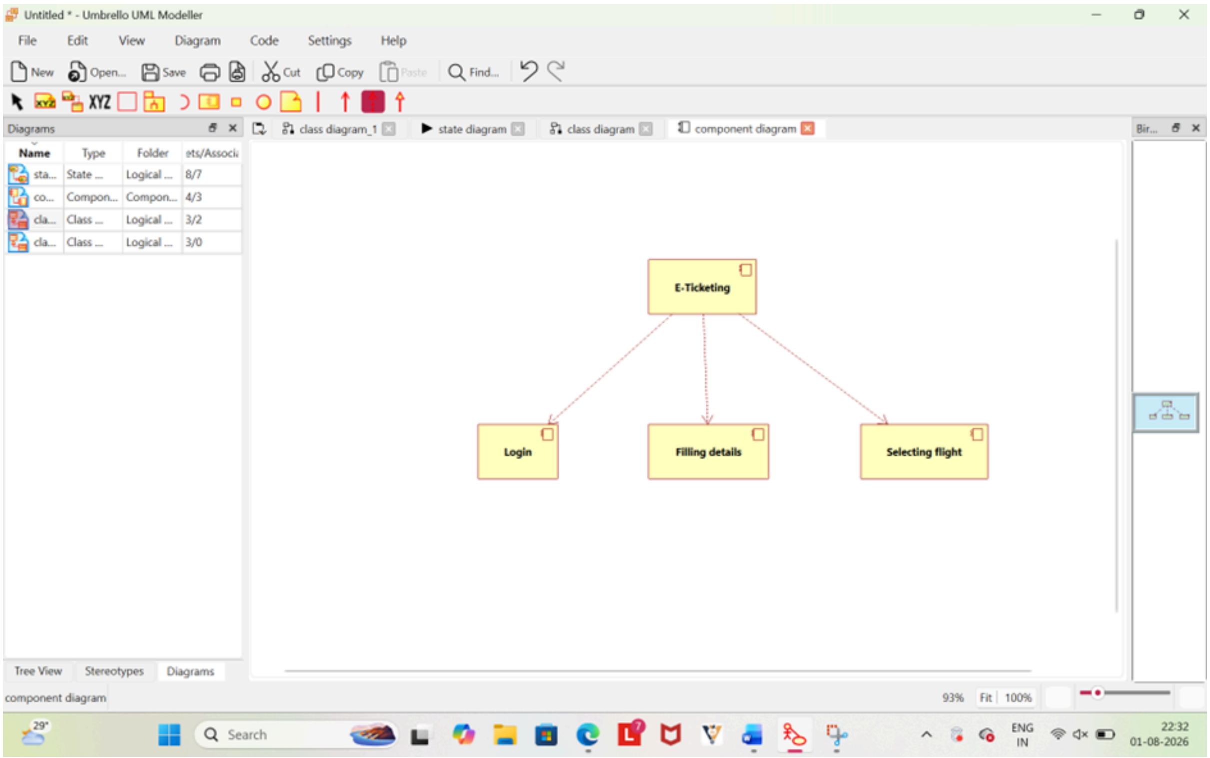


Class Diagram

6.

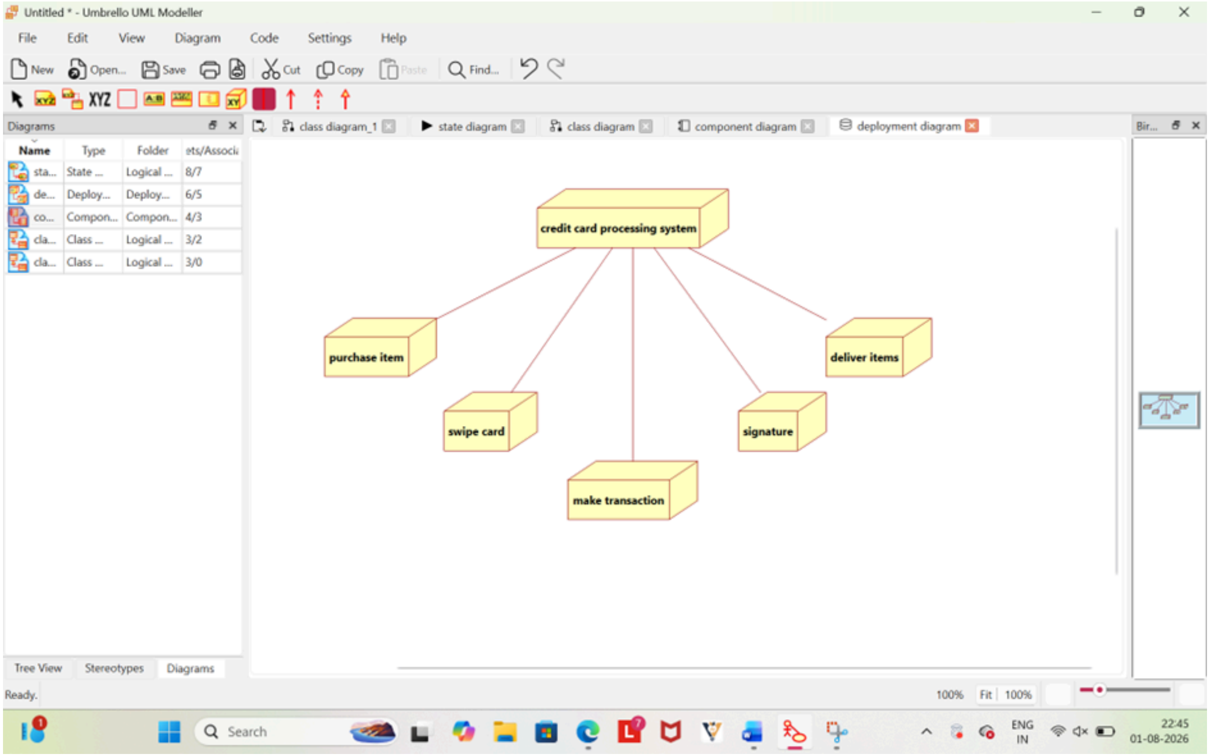


State diagram

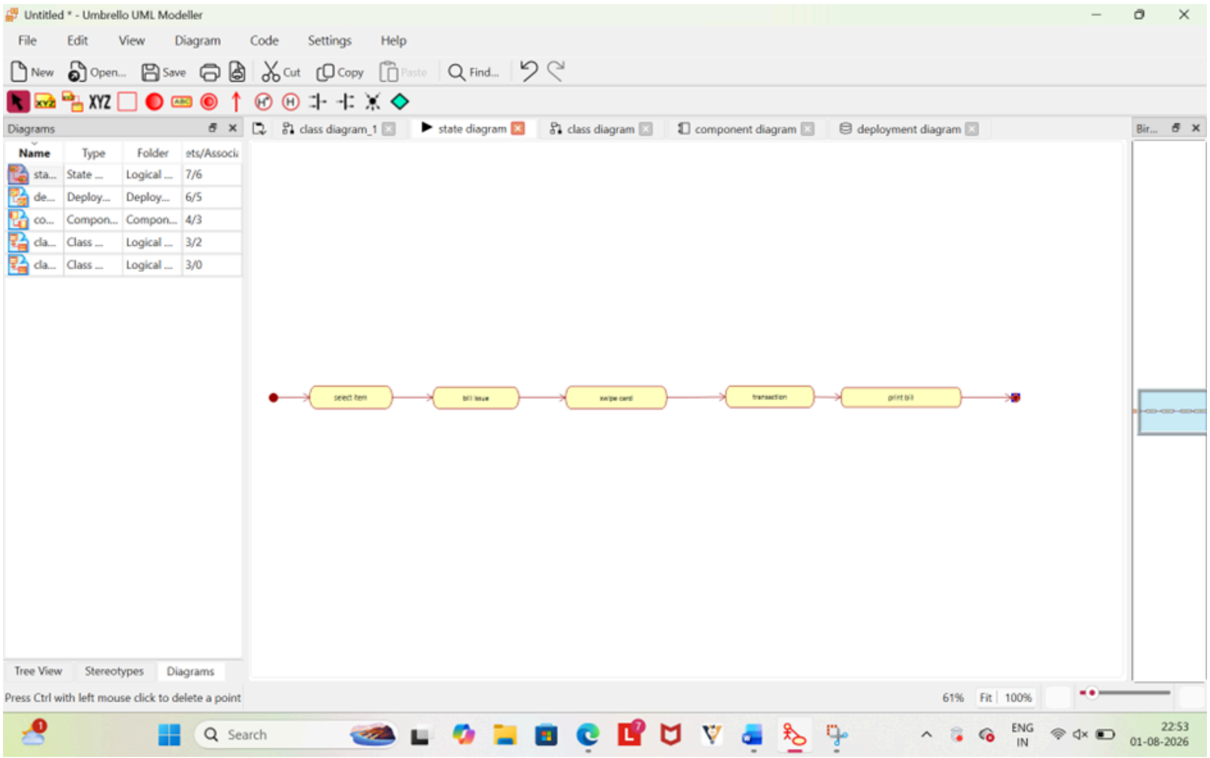


Component diagram

7.

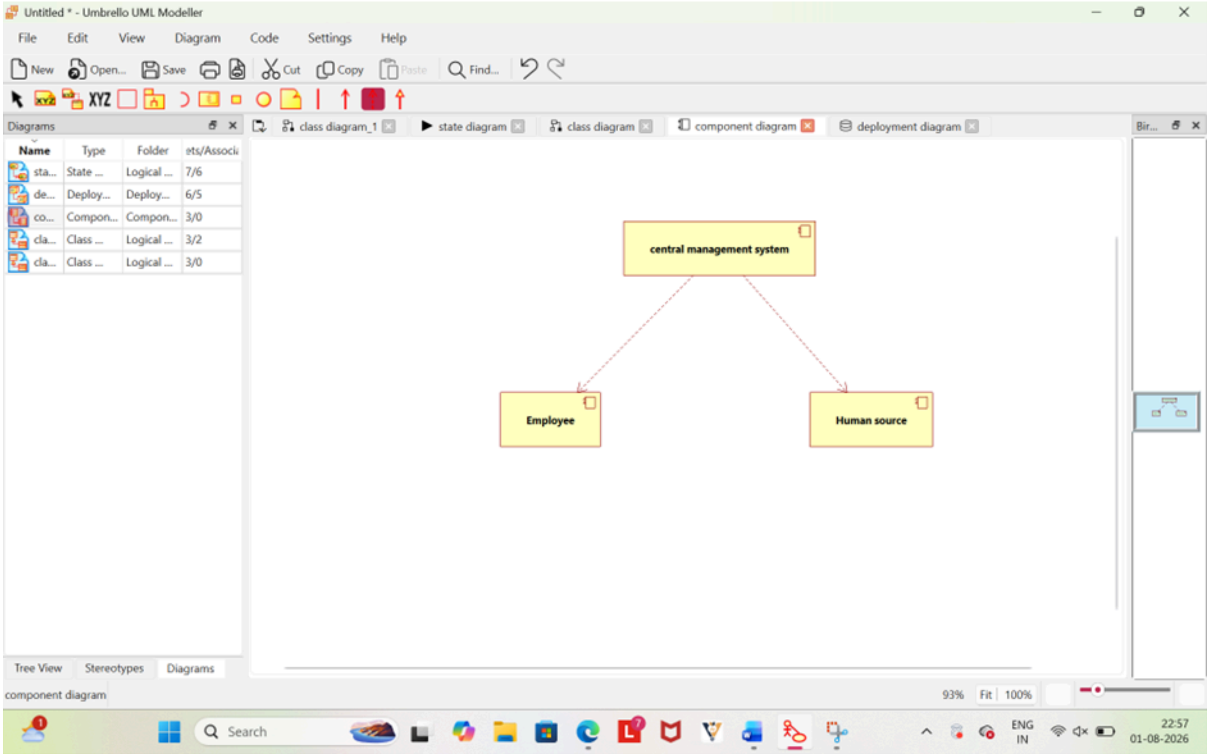


Deployment Diagram

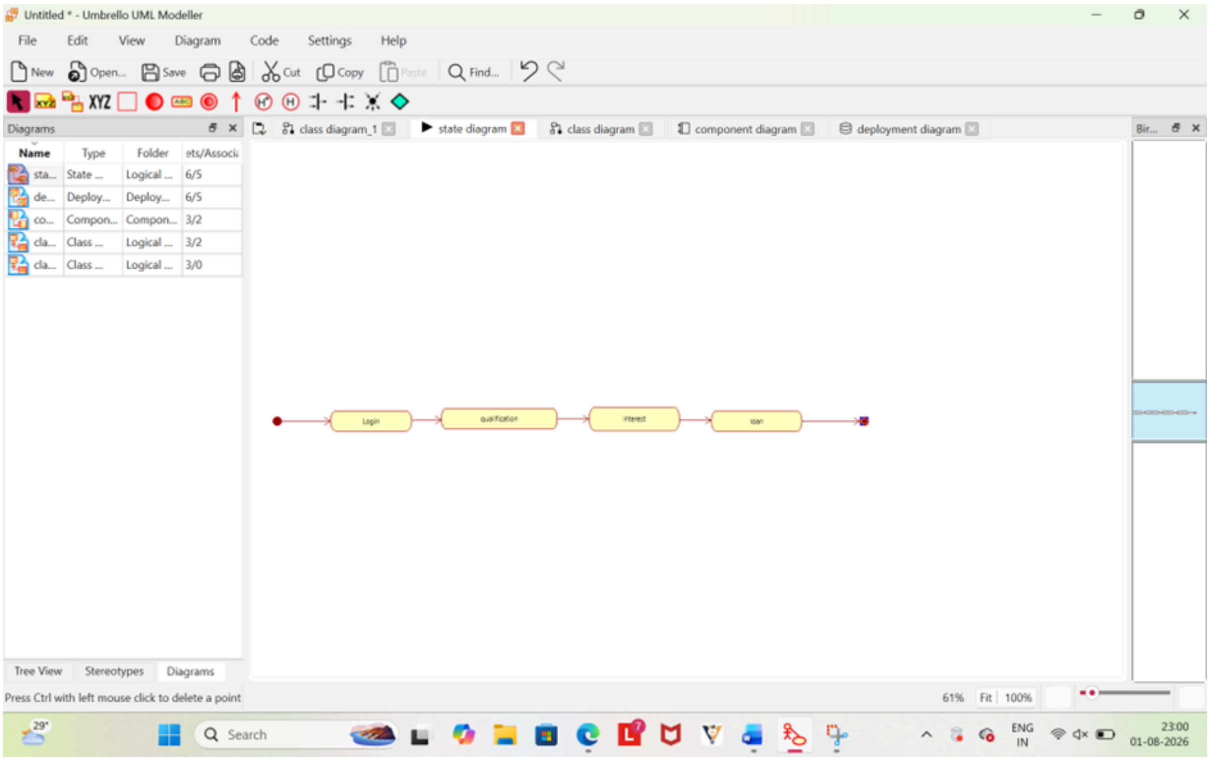


State Diagram

8.

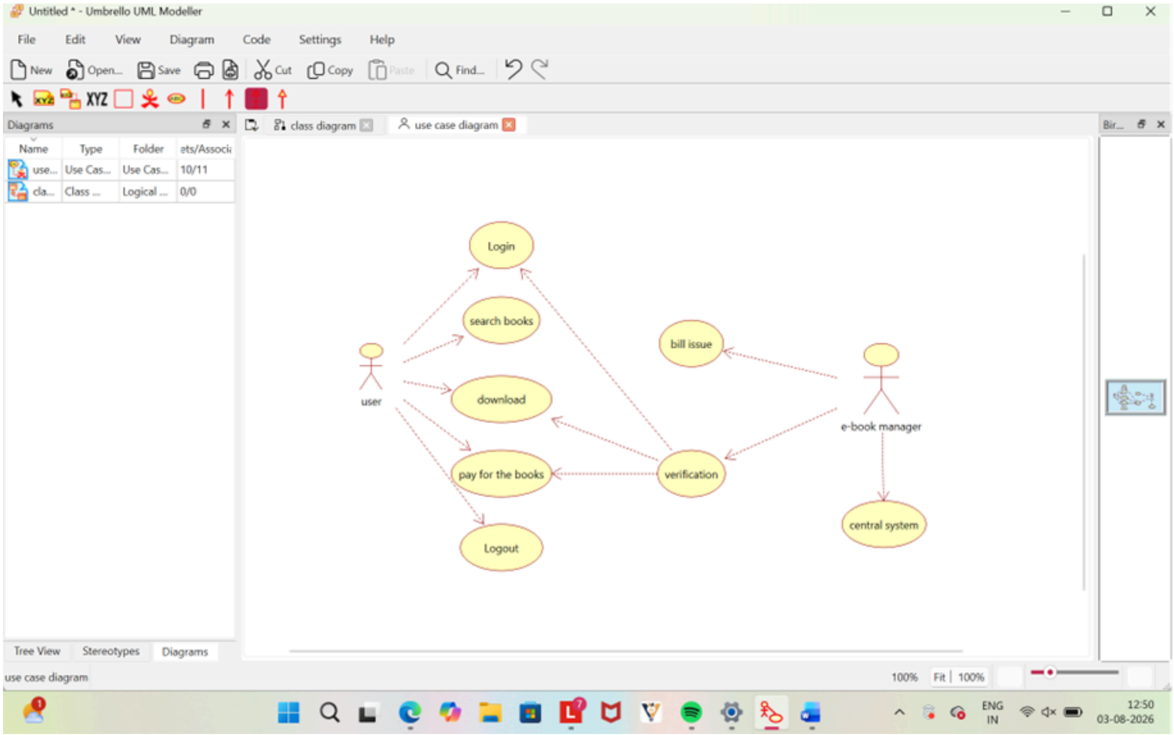


Component Diagram

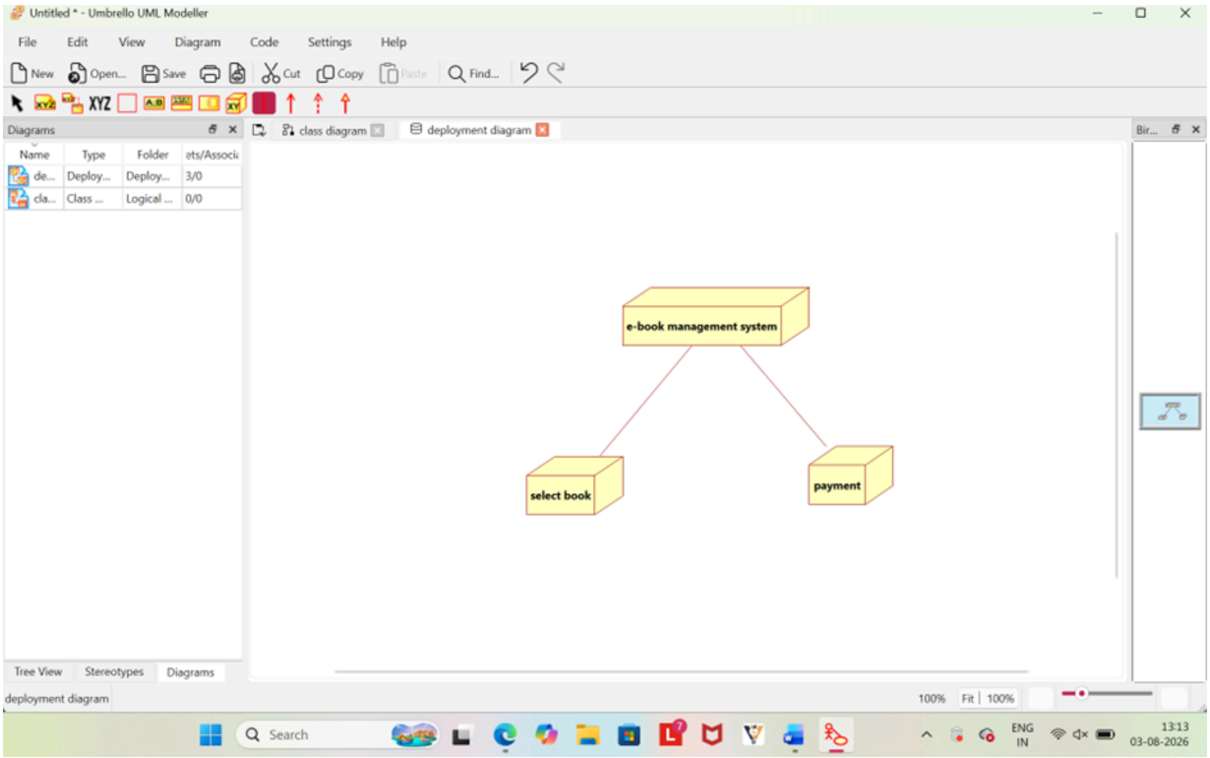


State Diagram

9.

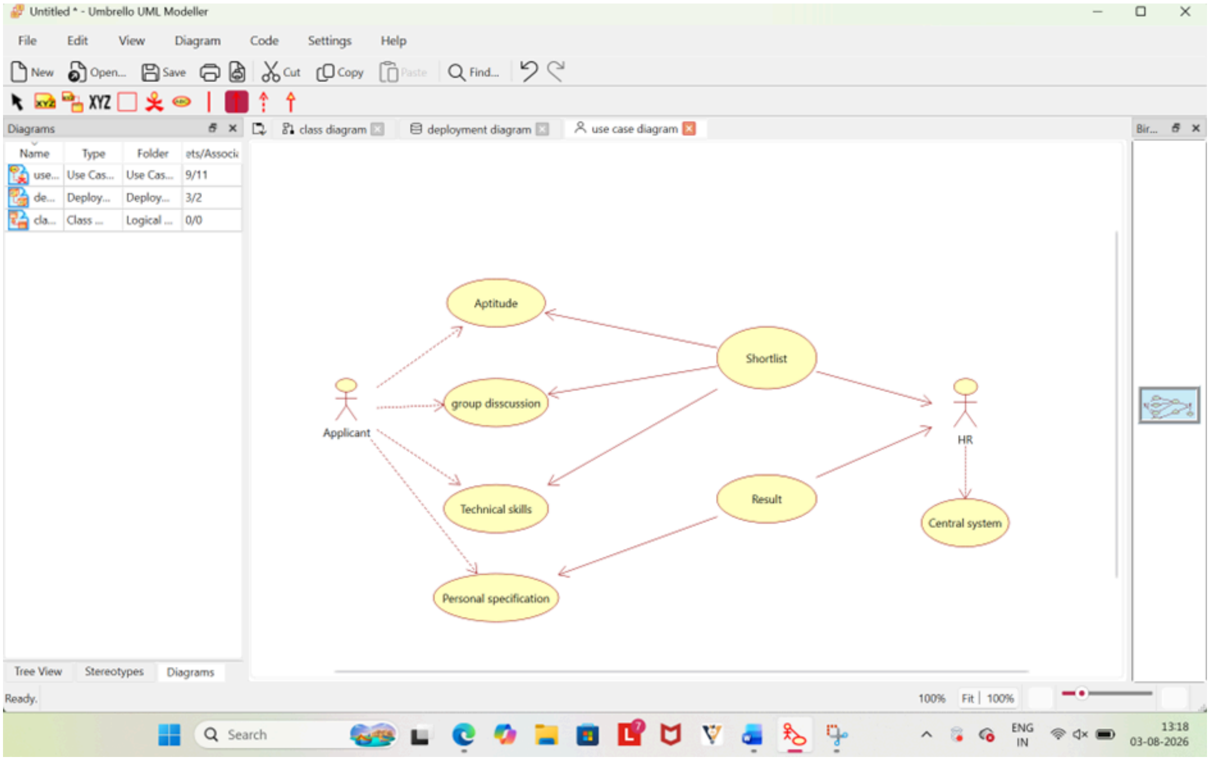


Use case diagram

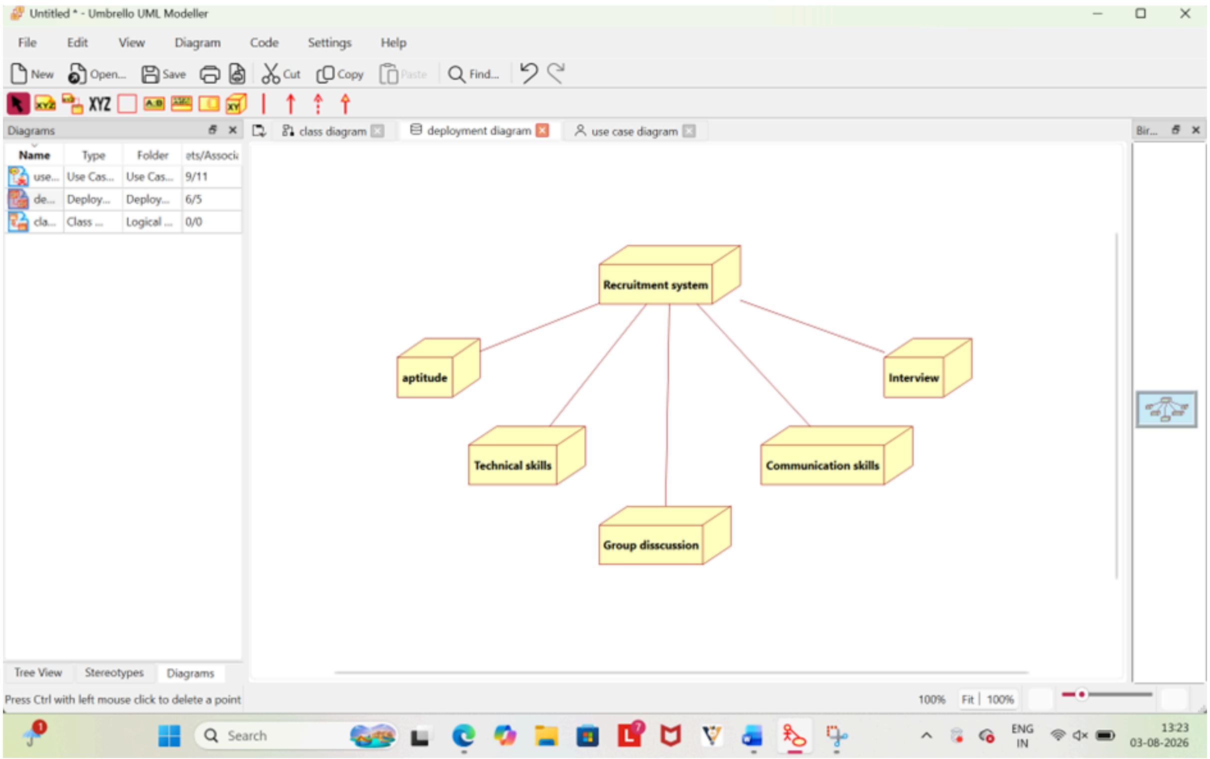


Deployment Diagram

10.



Use case Diagram



Deployment Diagram